

CUSUM Change-Point Detection Analysis

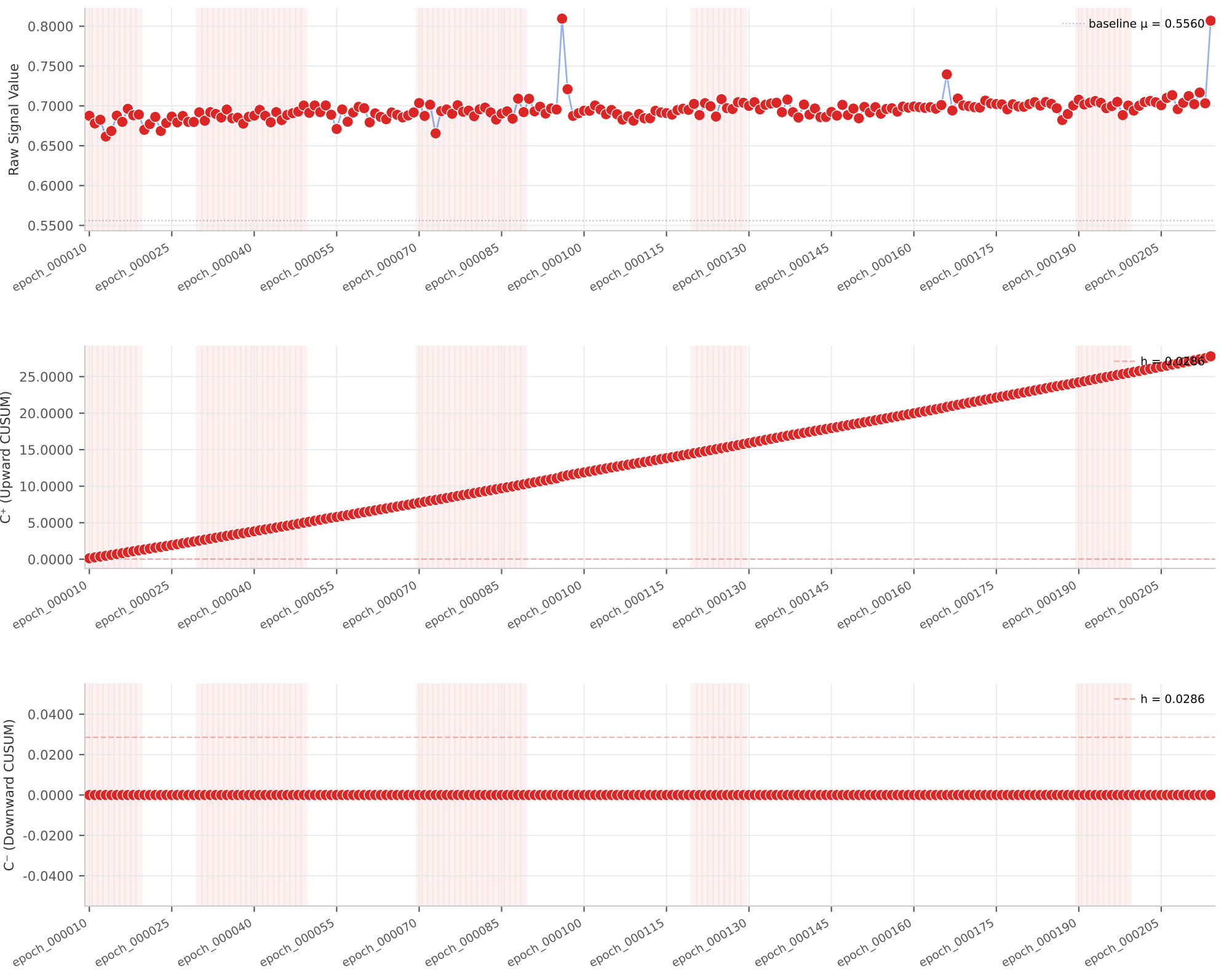
Cumulative Sum Control Chart | JS Divergence + New IP Ratio + Top-K Source

Run Parameters		Baseline CUSUM Parameters			Detection Summary (Any Signal)	
Dataset	txtD280_5.0_2.0_cusum_sparse.json	Baseline JS	$\mu=0.5560$	$\sigma=0.0071$	$k=0.0036$ $h=0.0286$	70
Timestamp	2026-02-23T14:49:08.964712	New IP Ratio	$\mu=0.0000$	$\sigma=0.0000$	$k=0.0000$ $h=0.0000$	135
Method	CUSUM Change-Point (JS + New IP + TopK)	Top-K	$\mu=0.0622$	$\sigma=0.0031$	$k=0.0015$ $h=0.0123$	0
Total epochs	—	False Negatives (FN)				0
Baseline epochs	— (?%)	Precision				0.341
Attack epochs	205	Recall				1.000
Malicious traffic	?%	F1 Score				0.509
CUSUM k-factor	0.5	Detection Rate				100.0%
CUSUM h-factor	4.0					
Mode	SPARSE					
Attack probability	0.3					
Window size	10					
Windows attacked	7					

Combined JS Divergence

CUSUM Signal Analysis | malicious = 0% | baseline = 5%

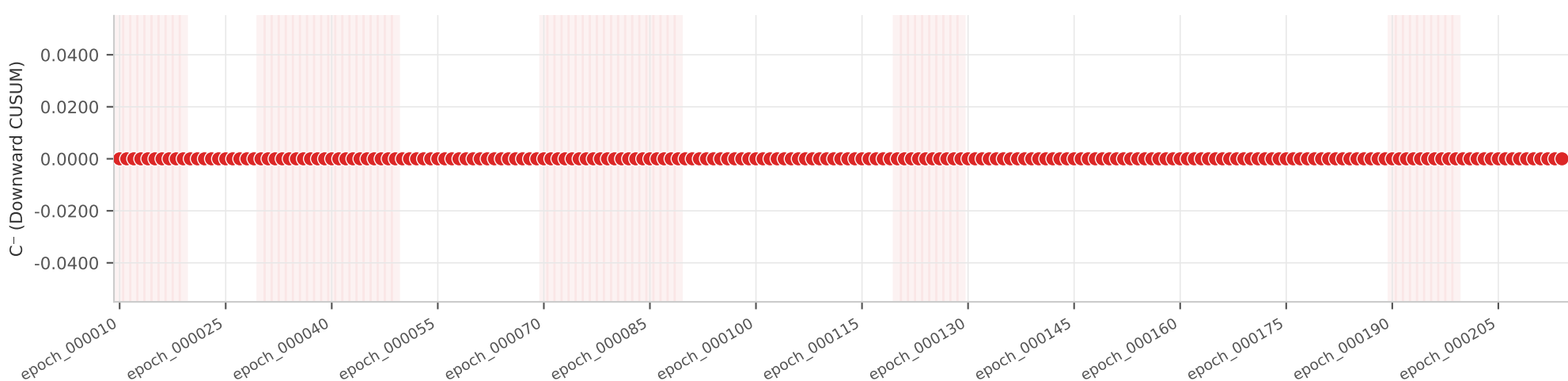
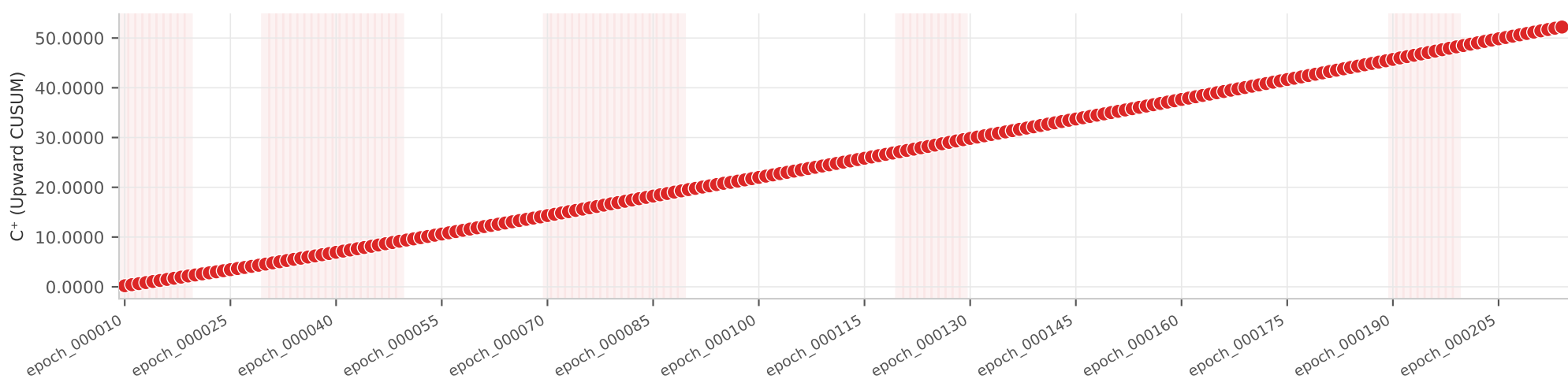
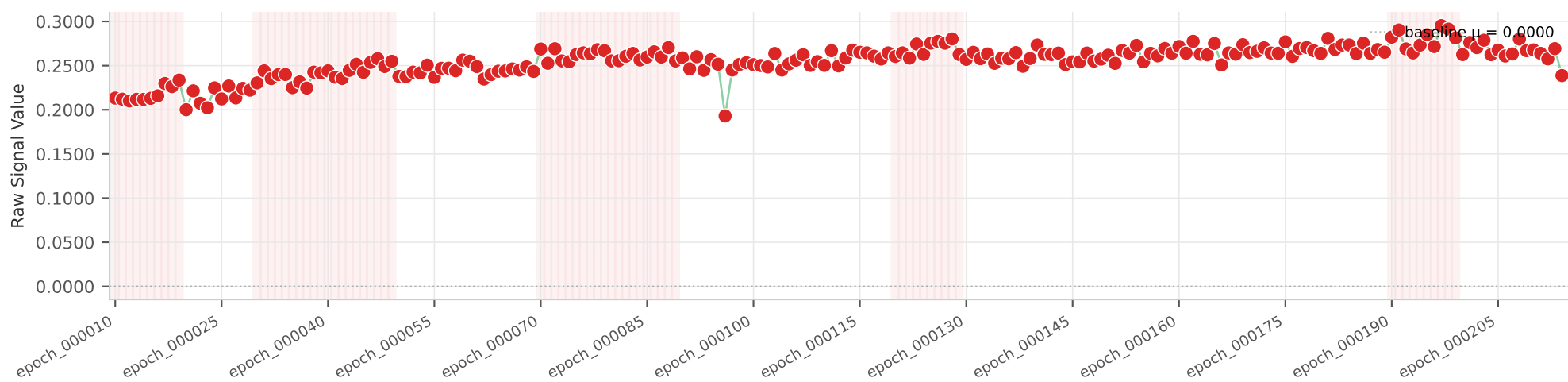
- CUSUM alarm fired
- Attacked epoch



New Source IP Ratio

- CUSUM alarm fired
- Attacked epoch

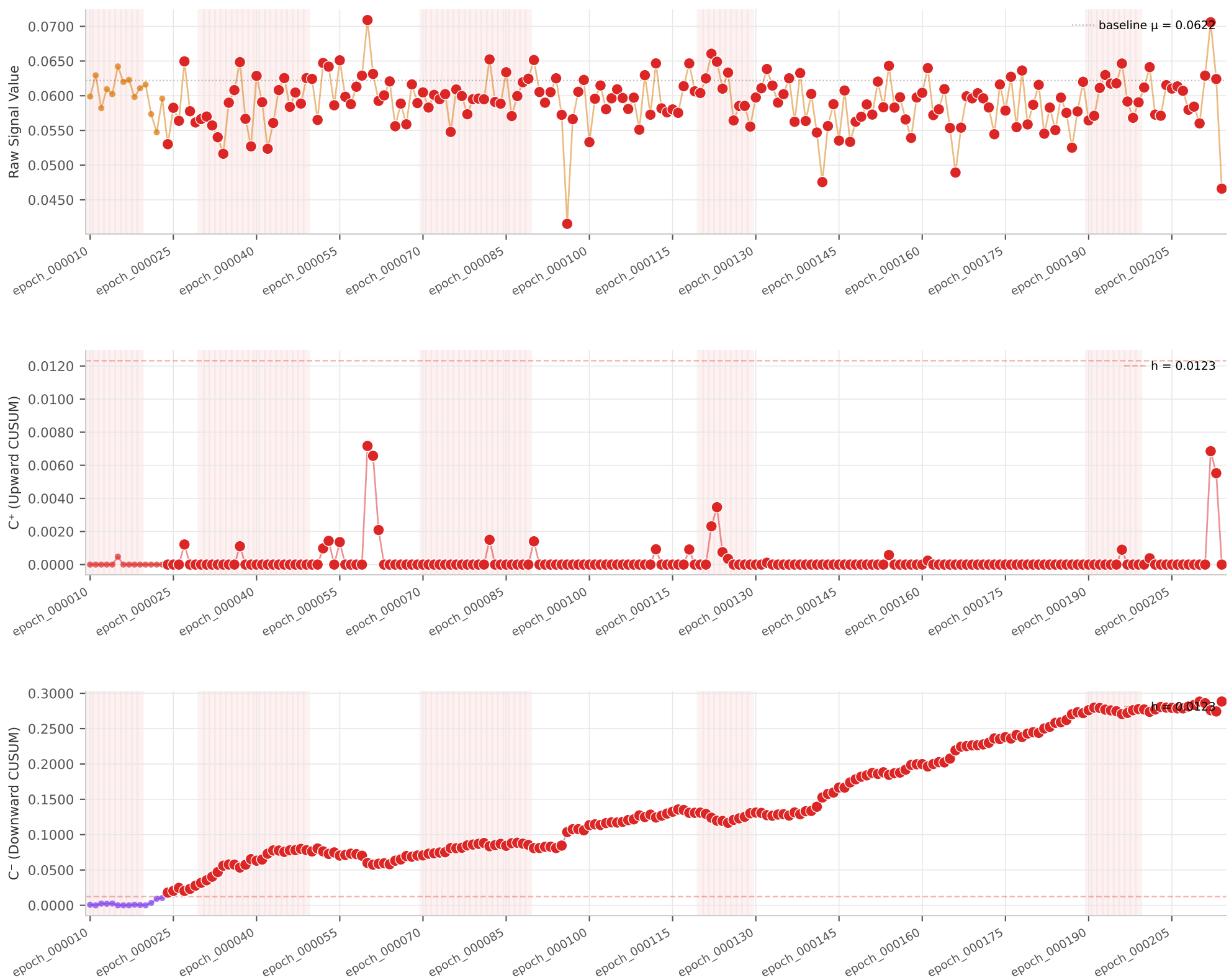
CUSUM Signal Analysis | malicious = 0% | baseline = 5%



Top-5 Source Concentration

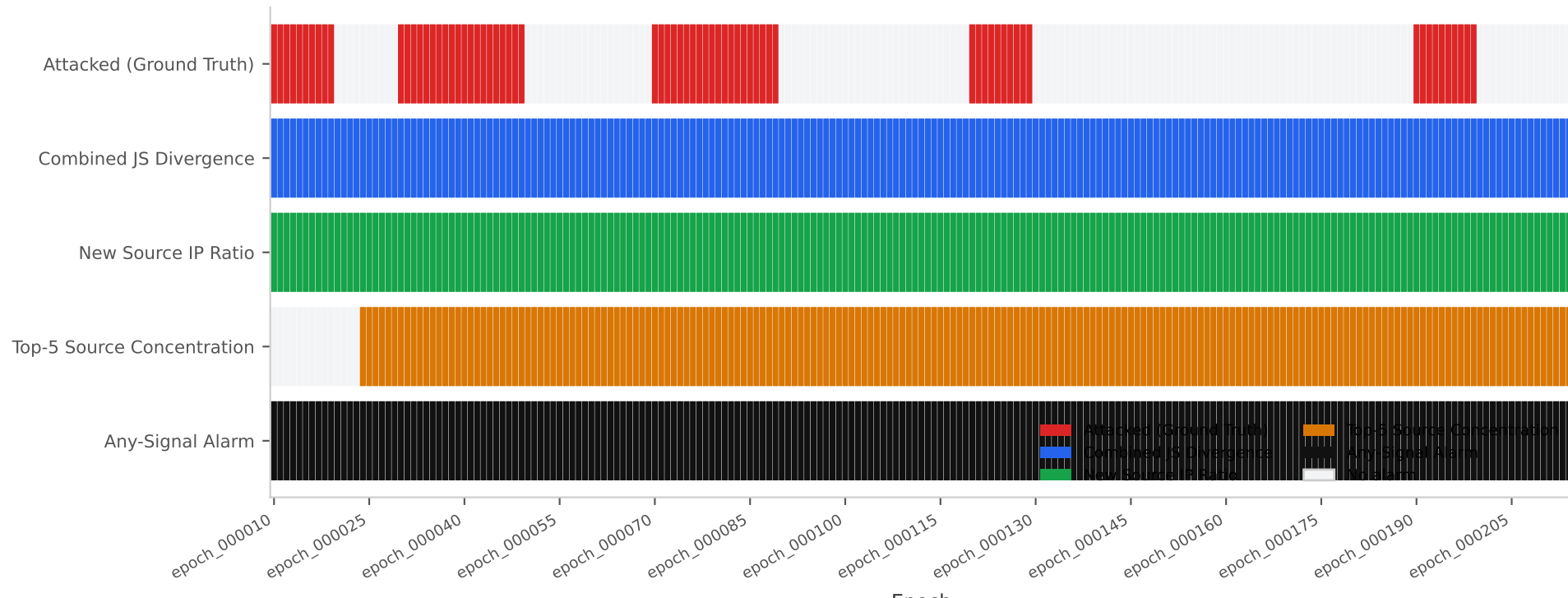
CUSUM Signal Analysis | malicious = 0% | baseline = 5%

- CUSUM alarm fired
- Attacked epoch



CUSUM Alarm Timeline

Filled cell = alarm fired for that epoch × signal



Detection Performance Metrics

Dataset: txtD280 | Malicious: ?% | Baseline: ?%

Metric	Value	Description
True Positives	70	Attacked epochs correctly alarmed
False Positives	135	Clean epochs incorrectly alarmed
True Negatives	0	Clean epochs correctly not alarmed
False Negatives	0	Attacked epochs missed
Precision	0.3415	$TP / (TP + FP)$
Recall	1.0000	$TP / (TP + FN)$
F1 Score	0.5091	Harmonic mean of Precision & Recall
Detection Rate	100.00%	Fraction of attacked epochs detected
False Alarm Rate	100.00%	Fraction of clean epochs with alarm